1. $A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 2 & 0 \\ 1 & -2 & -3 \end{pmatrix}$ $\qquad \kappa_\infty(A)$

2. Prager - Öttli:

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

$$|\Delta A| \leq \frac{1}{10} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \Delta b \leq \frac{1}{10} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

a) $\tilde{x}_1 = \begin{pmatrix} 2 \\ 0.7 \end{pmatrix}$ $\qquad$ b) $\tilde{x}_2 = \begin{pmatrix} 2.1 \\ 0.7 \end{pmatrix}$

(1) $A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 2 & 0 \\ 1 & -2 & -3 \end{pmatrix}$ $\qquad \kappa_\infty(A) = \|A\|_\infty \, \|A^{-1}\|_\infty$

$\|A\|_\infty = 6$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ -1 & 2 & 0 & 0 & 1 & 0 \\ 1 & -2 & -3 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & -2 & -3 & -1 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{3} & -\frac{1}{3} \end{array}\right)$$

$\|A^{-1}\|_\infty = 1$

$$\kappa_\infty(A) = \|A\|_\infty \, \|A^{-1}\|_\infty = 6 \cdot 1 = 6$$

② Prager - Öttli:

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

$$|\Delta A| \leq \underbrace{\frac{1}{10}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}}_{=B}, \qquad \Delta b \leq \underbrace{\frac{1}{10}\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{c}$$

a) $\tilde{x}_1 = \begin{pmatrix} 2 \\ 0.7 \end{pmatrix}$ b) $\tilde{x}_2 = \begin{pmatrix} 2.1 \\ 0.7 \end{pmatrix}$

$$|\tilde{r}| \overset{?}{\leq} B|\tilde{x}| + c, \qquad \tilde{r} = b - A\tilde{x}$$

a) $\tilde{x} = \tilde{x}_1 = \begin{pmatrix} 2 \\ 0.7 \end{pmatrix}$

$$\tilde{r} = b - A\tilde{x} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 0.7 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 3.3 \\ 2.7 \end{pmatrix}$$

$$\tilde{r} = \begin{pmatrix} -0.3 \\ 0.3 \end{pmatrix}$$

$$B|\tilde{x}| + c = \frac{1}{10}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 0.7 \end{pmatrix} + \frac{1}{10}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{10}\begin{pmatrix} 2.7 \\ 2.7 \end{pmatrix} + \frac{1}{10}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{10}\begin{pmatrix} 3.7 \\ 3.7 \end{pmatrix}$$

$$= \begin{pmatrix} 0.37 \\ 0.37 \end{pmatrix}$$

$$|\tilde{r}| \qquad\qquad B|\tilde{x}| + c$$

$$\begin{pmatrix} 0.3 \\ 0.3 \end{pmatrix} \leq \begin{pmatrix} 0.37 \\ 0.37 \end{pmatrix}$$

$\tilde{x}_1$ akzeptiert!

b) $\tilde{x} = \tilde{x}_2 = \begin{pmatrix} 2.1 \\ 0.7 \end{pmatrix}$

$\tilde{r} = \begin{pmatrix} -0.5 \\ \hline 0.2 \end{pmatrix}$, $B|\tilde{x}| + c = \begin{pmatrix} 0.38 \\ \hline 0.38 \end{pmatrix}$

$|\tilde{r}| \not\leq B|\tilde{x}| + c$

$\underline{x_2 \quad \text{abgelehnt}}$